VIOLET: A Cascade Electromechanical Cipher Machine

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Abstract

This paper presents a rigorous mathematical and cryptographic treatment of **Violet**, a novel cascade electromechanical cipher architecture that composes a reflectorless rotor permutation stage with an independent stepping-switch permutation stage, yielding a time-varying polyalphabetic substitution cipher over the symmetric group S_{26} of the 26-letter Latin alphabet. The system is formally modeled as a deterministic finite-state machine whose encryption function at each discrete time step is the composition of two independently evolving permutations drawn from S_{26}. A complete algebraic framework is developed within the theory of symmetric groups, cyclic groups, and wreath products, enabling formal proofs of structural properties including closure under composition, generic non-reciprocity of the cipher transformation, the existence of fixed points in accordance with classical derangement theory, and a period theorem establishing that the global period of the machine equals the least common multiple of the independent rotor and stepping-switch cycles. A full keyspace derivation is carried out, and for a canonical machine instance comprising five rotors, six stepping switches, and a ten-pair plugboard, the total keyspace is computed to exceed 2^{111} possible operational keys, yielding an effective key entropy of approximately 111 bits - substantially exceeding both the three-rotor Enigma ($\approx 2^{77}$) and the Japanese Purple machine ($\approx 2^{60}$). The statistical mixing behaviour of the composite cipher is analyzed through the lens of ergodic theory on finite groups, demonstrating that over sufficiently long message lengths the distribution of output permutations converges to approximate uniformity on S_{26}, with the rate of

convergence governed by the spectral gap of the associated random walk on the Cayley graph. A detailed cryptanalytic assessment is provided, demonstrating that the cascade architecture resists Bombe-style cribs, cycle-structure attacks, and depth attacks, while a formal meet-in-the-middle analysis establishes that the two-stage composition does not admit efficient decomposition attacks unless the attacker can enumerate an intermediate permutation space of order $26! \approx 4.03 \times 10^{26}$, rendering such attacks computationally infeasible. The mechanical realizability of the design using mid-twentieth-century electromechanical components - including Strowger stepping switches, wired rotor drums, and relay-driven logic - is discussed in detail, establishing Violet as a historically plausible cipher system that would have represented a significant advance over contemporaneous machines.

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1. Introduction

1.1 Historical Context

The history of cryptography in the twentieth century is inextricably linked to the development of electromechanical cipher machines. From the earliest commercial Hebern machines of the 1920s through the wartime deployment of the German Enigma, the American SIGABA, and the Japanese Purple, a central design challenge persisted: how to generate long, complex sequences of substitution alphabets from compact, portable mechanical devices. Two principal paradigms emerged.

The first paradigm, **rotor machines**, achieved polyalphabetic substitution by passing electrical signals through a cascade of wired rotary discs (rotors), each implementing a fixed permutation of the alphabet, with the rotors advancing in an odometer-like fashion to produce a time-varying composite permutation. The most famous exemplar is the Enigma machine, adopted by the German military in various configurations from the late 1920s onward. The Enigma employed three (later four) rotors selected from a larger set, a plugboard (Steckerbrett) providing an initial and final transposition, and a reflecting rotor (Umkehrwalze) that ensured the cipher was its

own inverse - a property termed **reciprocity** or **involutory** character. While operationally convenient, this reciprocal structure introduced a critical cryptographic weakness: no letter could encrypt to itself, providing a powerful crib for cryptanalysts at Bletchley Park and enabling the construction of the Bombe, an electromechanical device that exploited known-plaintext relationships to recover daily key settings.

The second paradigm, **stepping-switch machines**, replaced rotating wired discs with banks of telephone-exchange stepping switches (Strowger switches or similar selectors), which implemented permutations by electrically connecting input terminals to output terminals according to the current switch position. The most prominent example is the Japanese Type B cipher machine, codenamed **Purple** by American cryptanalysts. Purple divided its 26-letter alphabet into two groups - a set of six letters ("sixes") and a set of twenty letters ("twenties") - and applied separate permutation processes to each, driven by cascaded stepping switches with distinct periods. This partition, once identified by William Friedman's team at the Signal Intelligence Service, proved to be a devastating structural weakness, as it reduced the effective cryptanalytic search space dramatically.

Both paradigms produced machines that, despite their ingenuity, suffered from identifiable structural weaknesses. Enigma's reciprocity and its self-stenciling property (no letter encrypts to itself) provided entry points for known-plaintext attacks. Purple's alphabet partition and the relatively small state space of its six-letter group enabled complete reconstruction. These historical failures motivate the present investigation: **what cryptographic properties arise from composing both paradigms into a single, cascade cipher system?**

1.2 The Cascade Principle

The concept of cascading independent cipher stages has a natural algebraic interpretation. If ρ_t and σ_t are two independently evolving families of permutations in S_{26}, their composition $E_t = \sigma_t \circ \rho_t$ produces a third family whose algebraic and statistical properties are, in general, richer than those of either constituent. The cascade principle is well-established in modern block cipher design - the substitution-

permutation network (SPN) structure of the Advanced Encryption Standard (AES) is, in essence, a cascade of simple round transformations - but its application to electromechanical cipher machines has received comparatively little attention in the literature.

This paper introduces **Violet**, a hypothetical but historically plausible electromechanical cipher machine that realizes the cascade principle by combining a reflectorless rotor stage with an independent stepping-switch stage. The design deliberately avoids two of the most consequential weaknesses of its historical predecessors: the reflector of the Enigma (which enforces reciprocity and precludes self-encryption) and the alphabet partition of Purple (which reduces the effective permutation space).

1.3 Contributions and Outline

The contributions of this paper are as follows:

1. A **complete formal model** of the Violet cipher as a deterministic finite-state machine over the symmetric group S_{26}, with explicit algebraic representations of both the rotor and stepping-switch stages.
2. **Rigorous proofs** of seven structural theorems characterizing the closure, non-reciprocity, fixed-point distribution, periodicity, state-space evolution, temporal diffusion, and statistical mixing properties of the composite cipher.
3. A **full keyspace derivation and entropy calculation** for a canonical machine instance, yielding a key entropy exceeding 270 bits.
4. A **detailed cryptanalytic assessment** covering known-plaintext attacks, ciphertext-only attacks, Bombe-style crib attacks, cycle-structure attacks, depth attacks, and meet-in-the-middle decomposition attacks, with formal complexity analyses.
5. A **comparative analysis** situating Violet relative to Enigma and Purple in terms of keyspace, period, structural properties, and known weaknesses.
6. A **mechanical feasibility analysis** demonstrating that Violet could have been constructed using mid-twentieth-century electromechanical technology.

The remainder of this paper is organized as follows. Section 2 provides background on rotor machines and stepping-switch machines. Section 3 establishes the mathematical foundations. Section 4 defines the formal framework of time-varying permutation ciphers. Section 5 describes the Violet architecture in detail. Sections 6 and 7 present the rotor and stepping-switch stage models, respectively. Section 8 develops the state-space representation. Section 9 presents and proves seven structural theorems. Section 10 analyzes the period and cycle structure. Section 11 derives the keyspace and computes entropy. Section 12 examines statistical mixing properties. Section 13 addresses cryptanalytic considerations in depth. Section 14 presents the comparative analysis. Section 15 discusses mechanical feasibility. Section 16 examines security limitations from a modern perspective. Section 17 concludes. Mathematical appendices provide supplementary derivations.

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2. Background on Rotor Machines and Stepping-Switch Machines

2.1 Rotor Machines

A rotor machine encrypts plaintext by passing an electrical signal through a sequence of wired rotary discs. Each rotor implements a fixed permutation $R_i \in S_{26}$ determined by its internal wiring. When a rotor is displaced by s positions from its reference orientation, the effective permutation becomes the conjugate $\tau_s^{-1} \circ R_i \circ \tau_s$, where τ_s denotes the cyclic shift $x \mapsto x + s \pmod{26}$. The rotors advance according to a stepping mechanism - typically odometer-style in the Enigma, with a "double-stepping" anomaly caused by mechanical pawl interactions - producing a time-varying sequence of composite permutations.

The Enigma machine further incorporated a **reflector** (Umkehrwalze), a fixed-point-free involution $U \in S_{26}$ satisfying $U = U^{-1}$ and $U(x) \neq x$ for all x. The signal passed through the rotor stack, was reflected by U, and returned through the rotors in reverse order, yielding a composite cipher transformation:

$$E_t^{\text{Enigma}} = P \circ R_1(t) \circ R_2(t) \circ R_3(t) \circ U \circ R_3(t)^{-1} \circ R_2(t)^{-1} \circ R_1(t)^{-1} \circ P^{-1}$$

where P is the plugboard permutation. This construction ensures $E_t = E_t^{-1}$ - the encryption operation is its own inverse - which simplifies key management but introduces the cryptanalytically exploitable property that no plaintext letter can encrypt to itself.

The SIGABA (also known as ECM Mark II), developed by William Friedman and Frank Rowlett for the United States military, represented a more sophisticated rotor design. SIGABA employed 15 rotors organized into three banks: five cipher rotors, five control rotors, and five index rotors. The control and index rotors generated a pseudorandom stepping pattern for the cipher rotors, producing irregular, non-odometric advancement. SIGABA was never broken during the war.

2.2 Stepping-Switch Machines

Stepping-switch cipher machines replace the physical rotation of wired discs with the incremental advancement of electromechanical telephone switches. A Strowger stepping switch is a device with typically 25 positions (in a 25-point variant), where each position connects a set of input terminals to a set of output terminals according to a wiring pattern determined at manufacture. As the switch advances through its positions, the effective permutation changes.

The Japanese Type B cipher machine (Purple) employed three 25-point, six-level stepping switches to encipher 20 letters of the alphabet (the "twenties"), while the remaining six letters (the "sixes") were enciphered by a simpler mechanism equivalent to a single 25-point switch. The three switches for the twenties group were cascaded: the output of one fed the input of the next, and they advanced at different rates (one stepping with each character, one stepping every 25 characters, and one every 625 characters), producing a composite period of $25^3 = 15{,}625$ for the twenties group.

Purple's most consequential weakness was its rigid partition of the alphabet into the sixes and twenties groups. This partition remained static across all key settings and was identifiable through frequency analysis, since the six high-frequency vowels (A, E, I, O, U, Y in the most common configuration) were always enciphered together. Once American cryptanalysts identified this partition, the effective keyspace for each group could be attacked independently, reducing the problem from a 26-letter permutation search to two much smaller searches.

2.3 Motivation for a Hybrid Design

The historical record suggests several unrealized design possibilities. A hybrid machine combining the strengths of both paradigms - the high combinatorial complexity of rotor wiring with the flexible state evolution of stepping switches - while avoiding their respective weaknesses - the reflector-induced reciprocity of Enigma and the alphabet partition of Purple - would potentially yield a cryptographically stronger system. The Violet cipher is a concrete realization of this design philosophy.

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3. Mathematical Foundations

3.1 Alphabet and Symbol Space

Let the plaintext and ciphertext alphabets be the set

$$\Sigma = \{0, 1, 2, \ldots, 25\}$$

representing the 26 letters of the Latin alphabet under the standard identification $A \mapsto 0, B \mapsto 1, \ldots, Z \mapsto 25$. The set Σ is equipped with the structure of the cyclic group $\mathbb{Z}_{26} = \mathbb{Z}/26\mathbb{Z}$ under addition modulo 26.

3.2 The Symmetric Group S_{26}

The **symmetric group** S_{26} is the group of all bijections $\pi : \Sigma \to \Sigma$ under function composition. Its order is

$$|S_{26}| = 26! = 403{,}291{,}461{,}126{,}605{,}635{,}584{,}000{,}000 \approx 4.03 \times 10^{26}.$$

The group S_{26} is non-abelian for $|\Sigma| \geq 3$: there exist permutations $\alpha, \beta \in S_{26}$ such that $\alpha \circ \beta \neq \beta \circ \alpha$. This non-commutativity is fundamental to the cryptographic properties of cascade ciphers, as it ensures that the order of composition matters and that the inverse of a composition reverses the order of the constituent permutations.

Every permutation $\pi \in S_{26}$ admits a unique decomposition (up to ordering) into disjoint cycles. The **cycle type** of π is the partition $\lambda = (1^{a_1} 2^{a_2} \cdots 26^{a_{26}})$ of 26, where a_j denotes the number of j-cycles in the disjoint cycle decomposition of π. The cycle type determines the conjugacy class of π in S_{26} and controls many of its algebraic properties, including its order (the least common multiple of the cycle lengths) and its parity (even if the total number of even-length cycles is even, odd otherwise).

3.3 Cyclic Shift Operators

For each $k \in \mathbb{Z}_{26}$, define the **cyclic shift operator** $\tau_k : \Sigma \to \Sigma$ by

$$\tau_k(x) = x + k \pmod{26}.$$

The map $k \mapsto \tau_k$ is a group homomorphism from $(\mathbb{Z}_{26}, +)$ into S_{26}, and its image $\langle \tau_1 \rangle = \{\tau_0, \tau_1, \ldots, \tau_{25}\}$ is a cyclic subgroup of order 26. Conjugation by τ_k defines an inner automorphism of S_{26}; for any $\pi \in S_{26}$, the conjugate

$$\tau_k^{-1} \circ \pi \circ \tau_k$$

is the permutation obtained by "shifting the frame of reference" by k positions before applying π and then shifting back. This conjugation operation is the algebraic basis for the effect of rotor displacement.

3.4 Involutions and Plugboard Permutations

An **involution** in S_{26} is a permutation P satisfying $P^2 = \mathrm{id}$, equivalently $P = P^{-1}$. Every involution decomposes into disjoint transpositions and fixed points. A plugboard with m pairs implements an involution with m transpositions and $26 - 2m$ fixed points. The number of such involutions is

$$I(26, m) = \frac{26!}{(26-2m)!\ 2^m\ m!} = \binom{26}{2m} \cdot (2m-1)!!$$

where $(2m-1)!! = 1 \cdot 3 \cdot 5 \cdots (2m-1)$ is the double factorial. This counts the number of ways to select $2m$ letters from 26 and partition them into m unordered pairs.

3.5 Wreath Products and State Spaces

The state space of a rotor machine with r rotors, each having 26 angular positions, is naturally described as $\mathbb{Z}_{26}^r$. The stepping mechanism acts on this state space as a transformation $F : \mathbb{Z}_{26}^r \to \mathbb{Z}_{26}^r$. The algebraic structure of the set of permutations generated by all possible rotor states can be analyzed within the framework of wreath products. The **wreath product** $\mathbb{Z}_{26} \wr S_r$ acts on the state-indexed permutations, though a full wreath-product analysis is beyond the scope of the present work. We note, however, that the rotor stage generates a subgroup of S_{26} whose structure depends on the specific rotor wirings chosen.

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4. Formal Definition of Time-Varying Permutation Ciphers

4.1 Definition

A **time-varying permutation cipher** over alphabet Σ is a triple $(\mathcal{S}, F, G)$ where:

1. $\mathcal{S}$ is a finite set called the **state space**.

2. $F : \mathcal{S} \to \mathcal{S}$ is the **state transition function**, a deterministic map governing the evolution of the machine state.
3. $G : \mathcal{S} \to S_{26}$ is the **output function**, mapping each state to a permutation of Σ.

Given an initial state $s_0 \in \mathcal{S}$, the machine produces a sequence of states $s_0, s_1, s_2, \ldots$ via the recurrence $s_{t+1} = F(s_t)$, and the encryption function at time t is

$$E_t = G(s_t).$$

Encryption of plaintext symbol $x_t \in \Sigma$ yields ciphertext symbol

$$z_t = E_t(x_t) = G(s_t)(x_t).$$

Decryption is accomplished by

$$x_t = E_t^{-1}(z_t) = G(s_t)^{-1}(z_t)$$

which is well-defined since $G(s_t) \in S_{26}$ is a bijection.

4.2 Periodicity

Since $\mathcal{S}$ is finite and F is deterministic, the state sequence $\{s_t\}_{t \geq 0}$ is eventually periodic: there exist integers $\mu \geq 0$ (the **preperiod**) and $N \geq 1$ (the **period**) such that $s_{t+N} = s_t$ for all $t \geq \mu$. If the state transition function F is a bijection (which is the case for both the rotor and stepping-switch stages of Violet, as we shall show), then $\mu = 0$ and the sequence is purely periodic from the start.

4.3 Composition of Time-Varying Ciphers

If $(\mathcal{S}_1, F_1, G_1)$ and $(\mathcal{S}_2, F_2, G_2)$ are two time-varying permutation ciphers operating over the same alphabet Σ, their **cascade composition** is the cipher $(\mathcal{S}_1 \times \mathcal{S}_2, F_1 \times F_2, G)$ where

$$(F_1 \times F_2)(s_1, s_2) = (F_1(s_1), F_2(s_2))$$

and

$$G(s_1, s_2) = G_2(s_2) \circ G_1(s_1).$$

The cascade applies the first cipher's permutation, then the second. The state space of the composite cipher is the Cartesian product of the constituent state spaces, and the states evolve independently. This independence is the key structural feature that distinguishes Violet from machines like SIGABA, where the control rotors influence the stepping of the cipher rotors.

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5. Detailed Violet Architecture

5.1 Overview

The Violet cipher machine consists of the following components, arranged in signal-path order:

1. **Plugboard** (P): A patch panel implementing a fixed involutory permutation with m transposition pairs, connecting the keyboard to the rotor stage entry.
2. **Rotor stage**: A bank of r rotors $R_1, R_2, \ldots, R_r$, each a wired disc implementing a fixed permutation of Σ, with stepping positions advancing according to an odometer-type mechanism. **No reflector is used** - the signal passes through the rotors in one direction only.
3. **Stepping-switch stage**: A bank of k stepping switches $W_1, W_2, \ldots, W_k$, each a 25-point electromechanical selector, cascaded so that the composite switching network implements a second, independently evolving permutation.
4. **Output panel**: Displays the ciphertext letter.

The signal path is:

$$\text{Plaintext} \xrightarrow{P} \text{Rotor stage} \xrightarrow{\rho_t} \text{Switch stage} \xrightarrow{\sigma_t} \text{Ciphertext}$$

and the full encryption function at time t is

$$E_t = \sigma_t \circ \rho_t$$

where ρ_t includes the plugboard transformation.

5.2 Absence of Reflector

The deliberate omission of a reflector is a crucial design decision. In the Enigma, the reflector U ensures $E_t = E_t^{-1}$, which means an operator can decrypt a message simply by typing the ciphertext into a machine set to the same key. This operational convenience comes at a severe cryptographic cost: the cipher becomes an involution, and in particular, no letter can encrypt to itself ($E_t(x) \neq x$ for all x). This self-stenciling property was one of the most powerful tools available to Allied cryptanalysts.

By omitting the reflector, Violet produces cipher transformations that are **not** involutions in general. Decryption requires computing $E_t^{-1} = \rho_t^{-1} \circ \sigma_t^{-1}$, which is operationally more complex (requiring either a reverse-wired machine or precomputed inverse tables) but cryptographically far stronger: the cipher may map a letter to itself, and the encryption and decryption transformations are distinct.

5.3 Full Alphabet Operation

Unlike Purple, which partitions the alphabet into two independently processed groups, Violet applies **both** permutation stages to the **full 26-letter alphabet**. Every letter passes through every rotor and every stepping switch. This ensures that the effective permutation at each time step is drawn from the full symmetric group S_{26}, not from a product of smaller symmetric groups. The absence of alphabet partition eliminates the class of attacks that proved so effective against Purple.

6. Rotor Stage - Complete Mathematical Model

6.1 Components

The rotor stage comprises:

- A plugboard permutation $P \in S_{26}$, an involution with m transposition pairs.
- An ordered sequence of r rotors $R_1, R_2, \ldots, R_r$, each a fixed permutation $R_i \in S_{26}$.
- A rotor state vector $\mathbf{s}^{(\rho)}(t) = (s_1(t), s_2(t), \ldots, s_r(t)) \in \mathbb{Z}_{26}^r$, where $s_i(t)$ is the angular position of rotor i at time t.

6.2 Rotor Conjugation

When rotor R_i is displaced by $s_i(t)$ positions from its reference orientation, the effective permutation it implements is

$$R_i(t) = \tau_{s_i(t)}^{-1} \circ R_i \circ \tau_{s_i(t)} = \tau_{-s_i(t)} \circ R_i \circ \tau_{s_i(t)}.$$

This is a conjugation of R_i by the cyclic shift $\tau_{s_i(t)}$. Algebraically, displacement by $s_i(t)$ positions shifts the input by $+s_i(t)$, applies the base wiring R_i, and then shifts back by $-s_i(t)$.

Lemma 6.1. *For any $R_i \in S_{26}$ and any $s, s' \in \mathbb{Z}_{26}$:*

$$\tau_s^{-1} \circ R_i \circ \tau_s = \tau_{s'}^{-1} \circ R_i \circ \tau_{s'} \iff s \equiv s' \pmod{26}$$

provided R_i does not commute with all cyclic shifts (i.e., $R_i \notin \langle \tau_1 \rangle$).

Proof. Suppose $\tau_s^{-1} R_i \tau_s = \tau_{s'}^{-1} R_i \tau_{s'}$. Then $\tau_{s'-s}^{-1} R_i \tau_{s'-s} = R_i$, i.e., R_i commutes with $\tau_{s'-s}$. The centralizer of $\langle \tau_1 \rangle$ in S_{26} is $\langle \tau_1 \rangle$ itself (since τ_1 is a 26-cycle and its centralizer in S_{26} is the cyclic group it generates). Therefore, if $R_i \notin \langle \tau_1 \rangle$, we must have $\tau_{s'-s} = \mathrm{id}$, hence $s' \equiv s \pmod{26}$. $\square$

This lemma confirms that each of the 26 angular positions of a rotor produces a **distinct** effective permutation (provided the rotor wiring is not itself a cyclic shift, which would be a degenerate and cryptographically worthless choice).

6.3 Composite Rotor Permutation

The composite rotor-stage permutation at time t is

$$\rho_t = R_r(t) \circ R_{r-1}(t) \circ \cdots \circ R_1(t) \circ P$$

$$= \left(\prod_{i=r}^{1} \tau_{-s_i(t)} \circ R_i \circ \tau_{s_i(t)} \right) \circ P$$

where the product notation denotes composition from right to left in decreasing index order.

6.4 Rotor Stepping Mechanism

The rotor state vector evolves according to an **odometer-type** stepping rule. Define the stepping function $F_\rho : \mathbb{Z}_{26}^r \to \mathbb{Z}_{26}^r$ as follows. Before each encryption, the rightmost rotor (rotor 1) advances by one position. When rotor i completes a full revolution (its position returns to 0), rotor $i+1$ advances by one position. Formally:

$$s_1(t+1) = s_1(t) + 1 \pmod{26}$$

For $i = 2, 3, \ldots, r$:

$$s_i(t+1) = \begin{cases} s_i(t) + 1 \pmod{26} & \text{if } s_{i-1}(t+1) = 0 \\ s_i(t) & \text{otherwise} \end{cases}$$

This is equivalent to interpreting $\mathbf{s}^{(\rho)}(t)$ as a base-26 number and incrementing it by 1 at each step. The period of this odometer is

$$N_\rho = 26^r$$

since the state vector cycles through all 26^r elements of $\mathbb{Z}_{26}^r$ before returning to its initial configuration.

Remark. The Violet design employs a clean odometer mechanism without the "double-stepping" anomaly present in the Enigma. In the three-rotor Enigma, mechanical pawl interactions cause the middle rotor to step twice in succession under certain conditions, reducing the effective period from $26^3 = 17{,}576$ to $26 \times 25 \times 26 = 16{,}900$. The Violet rotor stage avoids this anomaly by design, achieving the full theoretical period of 26^r.

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7. Stepping-Switch Stage - Complete Mathematical Model

7.1 Components

The stepping-switch stage comprises k stepping switches $W_1, W_2, \ldots, W_k$. Each switch W_j has 25 positions (labeled $0, 1, \ldots, 24$, corresponding to the positions of a standard Strowger selector), and at each position implements a fixed permutation of Σ. Denote the permutation implemented by switch W_j at position p as $W_j^{(p)} \in S_{26}$.

7.2 Switch State Vector

The state of the stepping-switch stage at time t is the vector

$$\mathbf{s}^{(\sigma)}(t) = (q_1(t), q_2(t), \ldots, q_k(t)) \in \mathbb{Z}_{25}^k$$

where $q_j(t) \in \{0, 1, \ldots, 24\}$ is the position of switch W_j at time t.

7.3 Switch Stepping Mechanism

The stepping-switch stage employs a cascaded stepping mechanism analogous to but distinct from the rotor odometer. Switch W_1 advances by one position with each character. When W_1 completes a full cycle (returns to position 0), W_2 advances by one position, and so on. Formally:

$$q_1(t+1) = q_1(t) + 1 \pmod{25}$$

For $j = 2, 3, \ldots, k$:

$$q_j(t+1) = \begin{cases} q_j(t) + 1 \pmod{25} & \text{if } q_{j-1}(t+1) = 0 \\ q_j(t) & \text{otherwise} \end{cases}$$

The period of this odometer on $\mathbb{Z}_{25}^k$ is

$$N_\sigma = 25^k.$$

7.4 Composite Switch Permutation

At time t, the composite permutation implemented by the stepping-switch stage is

$$\sigma_t = W_k^{(q_k(t))} \circ W_{k-1}^{(q_{k-1}(t))} \circ \cdots \circ W_1^{(q_1(t))}.$$

Each of the 25^k possible switch state vectors yields a (generally distinct) composite permutation in S_{26}.

7.5 Permutation Generation Quality

Unlike the rotor stage, where permutations are generated by conjugation of fixed wirings and thus have a structured algebraic relationship to one another, the stepping-switch stage can produce algebraically unrelated permutations at successive positions, since each switch position is independently wired. This provides an additional source

of cryptographic strength: the successive permutations σ_t and σ_{t+1} need not bear any simple algebraic relationship, frustrating attacks that attempt to exploit structured transitions between successive cipher alphabets.

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8. State-Space Representation of the Machine

8.1 Combined State Vector

The complete state of the Violet machine at time t is the combined vector

$$\mathbf{s}(t) = (\mathbf{s}^{(\rho)}(t),\ \mathbf{s}^{(\sigma)}(t)) \in \mathbb{Z}_{26}^r \times \mathbb{Z}_{25}^k = \mathcal{S}.$$

The total number of machine states is

$$|\mathcal{S}| = 26^r \cdot 25^k.$$

8.2 State Transition Function

The combined state transition function $F : \mathcal{S} \to \mathcal{S}$ is defined componentwise:

$$F(\mathbf{s}^{(\rho)}, \mathbf{s}^{(\sigma)}) = (F_\rho(\mathbf{s}^{(\rho)}),\ F_\sigma(\mathbf{s}^{(\sigma)}))$$

where F_ρ and F_σ are the odometer stepping functions defined in Sections 6.4 and 7.3, respectively. Since both F_ρ and F_σ are bijections on their respective state spaces (each is a cyclic permutation of a finite set), F is a bijection on $\mathcal{S}$.

8.3 Output Function

The output function $G : \mathcal{S} \to S_{26}$ maps each machine state to the corresponding encryption permutation:

$$G(\mathbf{s}^{(\rho)}, \mathbf{s}^{(\sigma)}) = \sigma(\mathbf{s}^{(\sigma)}) \circ \rho(\mathbf{s}^{(\rho)})$$

Theorem 8.1 (State-Space Evolution). *The Violet machine is a deterministic finite-state transducer. Given an initial state* $\mathbf{s}(0) \in \mathcal{S}$ *and a plaintext sequence* $(x_0, x_1, x_2, \ldots)$*, the ciphertext sequence* $(z_0, z_1, z_2, \ldots)$ *is uniquely determined by*

$$z_t = G(\mathbf{s}(t))(x_t), \quad \mathbf{s}(t+1) = F(\mathbf{s}(t))$$

and is independent of the plaintext symbols (the state evolution is autonomous).

Proof. The state transition function F depends only on the current state $\mathbf{s}(t)$ and not on the plaintext symbol x_t. Therefore the state trajectory $\{\mathbf{s}(t)\}_{t \geq 0}$ is fully determined by the initial state $\mathbf{s}(0)$ and is independent of the plaintext. The ciphertext at each step is then a deterministic function of the state and the plaintext symbol: $z_t = G(\mathbf{s}(t))(x_t)$. $\square$

Remark. The autonomy of state evolution - the fact that the stepping does not depend on the plaintext - is a common feature of classical cipher machines (including Enigma and Purple) and distinguishes them from modern stream ciphers with feedback modes.

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9. Structural Theorems

Theorem 9.1 (Closure)

For every time step $t \geq 0$*, the Violet encryption function* E_t *is a permutation of* Σ*:*

$$E_t \in S_{26}.$$

Proof. The plugboard permutation $P \in S_{26}$. Each rotor permutation $R_i \in S_{26}$, and conjugation by $\tau_{s_i(t)}$ preserves membership in S_{26} (since conjugation is an automorphism of S_{26}). Therefore $R_i(t) = \tau_{s_i(t)}^{-1} \circ R_i \circ \tau_{s_i(t)} \in S_{26}$. The composition $\rho_t = R_r(t) \circ \cdots \circ R_1(t) \circ P$ is a composition of elements of S_{26} and hence lies in S_{26}.

Similarly, each $W_j^{(q_j(t))} \in S_{26}$, so $\sigma_t \in S_{26}$. Finally, $E_t = \sigma_t \circ \rho_t \in S_{26}$. □

Theorem 9.2 (Non-Reciprocity)

For generic choices of rotor wirings and switch wirings, the Violet encryption function is not an involution: $E_t \neq E_t^{-1}$ for generic t.

Proof. Suppose $E_t = E_t^{-1}$. Then $E_t^2 = \mathrm{id}$, so $\sigma_t \circ \rho_t \circ \sigma_t \circ \rho_t = \mathrm{id}$, requiring $\sigma_t \circ \rho_t \circ \sigma_t = \rho_t^{-1}$, i.e., σ_t must conjugate ρ_t to its inverse. The set of pairs $(\rho, \sigma) \in S_{26} \times S_{26}$ satisfying this relation is a proper subset of $S_{26} \times S_{26}$. For independently chosen wiring configurations, the probability that this relation holds at any given t is negligible. Thus, for generic wirings, E_t is not an involution. □

Corollary 9.3. *Unlike the Enigma, the Violet cipher can encrypt a letter to itself.*

Theorem 9.4 (Fixed Point Distribution)

Let $X_t = |\{x \in \Sigma : E_t(x) = x\}|$ denote the number of fixed points of E_t. If E_t is modeled as a uniformly random permutation in S_{26}, then:

$$(i)\ \mathbb{E}[X_t] = 1 \qquad (ii)\ \mathrm{Var}(X_t) = 1 \qquad (iii)\ \Pr(X_t = j) \to \frac{e^{-1}}{j!} \text{ as } |\Sigma| \to \infty.$$

Proof. Define indicator variables $I_x = \mathbf{1}[E_t(x) = x]$ for each $x \in \Sigma$. Then $X_t = \sum_x I_x$.

(i) By linearity: $\mathbb{E}[X_t] = \sum_{x=0}^{25} \Pr(E_t(x) = x) = 26 \cdot \frac{1}{26} = 1$.

(ii) For the variance:
$\mathbb{E}[X_t^2] = \sum_x \mathbb{E}[I_x] + \sum_{x \neq y} \Pr(\pi(x) = x, \pi(y) = y) = 1 + 26 \cdot 25 \cdot \frac{1}{26 \cdot 25} = 2$. Thus $\mathrm{Var}(X_t) = 2 - 1 = 1$.

(iii) The count of permutations in S_n with exactly j fixed points is $\binom{n}{j} D_{n-j}$, where $D_m = m! \sum_{i=0}^{m} (-1)^i / i!$. As $n \to \infty$, $\Pr(X = j) \to e^{-1}/j!$, a Poisson distribution with parameter 1. □

Theorem 9.5 (Machine Period)

The period of the Violet machine is

$$N = \mathrm{lcm}(N_\rho, N_\sigma) = \mathrm{lcm}(26^r, 25^k).$$

Since $\gcd(26, 25) = 1$*, we have* $\gcd(26^r, 25^k) = 1$*, and therefore*

$$N = 26^r \cdot 25^k = |\mathcal{S}|.$$

Proof. The state transition $F = F_\rho \times F_\sigma$ acts on $\mathcal{S} = \mathbb{Z}_{26}^r \times \mathbb{Z}_{25}^k$. Since F_ρ has period $N_\rho = 26^r$ and F_σ has period $N_\sigma = 25^k$, the combined function has period $\mathrm{lcm}(N_\rho, N_\sigma)$. Since $26^r = 2^r \cdot 13^r$ and $25^k = 5^{2k}$ share no prime factors, $\gcd(26^r, 25^k) = 1$, so $\mathrm{lcm}(26^r, 25^k) = 26^r \cdot 25^k = |\mathcal{S}|$. The machine visits every state before repeating. □

Theorem 9.6 (Temporal Diffusion)

The ciphertext symbol z_t *depends on the complete initial state* $\mathbf{s}(0) \in \mathcal{S}$ *and the plaintext symbol* x_t*. For generic rotor and switch wirings, the function* $\mathbf{s}(0) \mapsto G(F^t(\mathbf{s}(0)))(x_t)$ *is a non-trivial function of all* $r + k$ *components of* $\mathbf{s}(0)$*.*

Proof. At time t, the machine state is $\mathbf{s}(t) = F^t(\mathbf{s}(0))$. Each rotor position $s_i(t)$ is a function of $s_i(0)$ and the carry propagation pattern depending on $s_1(0), \ldots, s_{i-1}(0)$. Similarly for switch positions. The encryption permutation E_t depends on all these positions through conjugations and switch lookups. Since each position produces a distinct permutation (Lemma 6.1), E_t is generically a non-trivial function of every component. □

Theorem 9.7 (Statistical Mixing - Ergodic Formulation)

Let $\mathcal{E} = \{E_0, E_1, \ldots, E_{N-1}\}$ *be the sequence of* $N = 26^r \cdot 25^k$ *permutations produced by one full period of the Violet machine. For any subset* $S \subseteq \Sigma$ *and any input* $x \in \Sigma$*:*

$$\frac{1}{N}\sum_{t=0}^{N-1}\mathbf{1}[E_t(x)\in S]\to\frac{|S|}{26}$$

as $r+k\to\infty$, where convergence is in the sense that the deviation from $|S|/26$ tends to zero.

Proof sketch. The sequence $\{E_t\}$ executes a deterministic walk on S_{26} driven by the state evolution. As the number of independent components (r rotors and k switches) increases, the effective permutation E_t becomes a product of an increasing number of independently varying factors. By results on mixing times for random walks on symmetric groups (Diaconis and Shahshahani, 1981), the distribution of a product of $\Theta(\log n)$ or more independent, non-degenerate permutations converges rapidly to the uniform distribution on S_n in total variation distance. While the Violet walk is deterministic rather than random, the large number of independently chosen wiring tables ensures that the sequence $\{E_t\}$ explores a large and well-distributed subset of S_{26}. The time average of $\mathbf{1}[E_t(x)\in S]$ converges to the space average $|S|/26$ as the orbit length grows, by an equidistribution argument analogous to the ergodic theorem for cyclic group actions. □

Remark. We emphasize that this is an asymptotic and heuristic result, not a rigorous proof that the output distribution is exactly uniform. The actual distribution depends on the specific wiring tables. The claim is that the cascade architecture promotes rapid approach to uniformity, in contrast to single-stage machines where structural regularities (such as the Enigma reflector) prevent uniform mixing.

10. Period and Cycle Analysis

10.1 Individual Component Periods

The rotor stage, operating as an odometer on $\mathbb{Z}_{26}^r$, has period $N_\rho = 26^r$. Each individual rotor completes a full revolution every 26 keystrokes; rotor i completes a full revolution every 26^i keystrokes.

The stepping-switch stage has period $N_\sigma = 25^k$. Switch W_j completes a full cycle every 25^j keystrokes.

10.2 Coprimality and Maximal Period

Since $\gcd(26, 25) = 1$, by the Chinese Remainder Theorem the combined odometer on $\mathbb{Z}_{26}^r \times \mathbb{Z}_{25}^k$ is isomorphic to a cyclic group of order $26^r \cdot 25^k$, and the state trajectory is a single cycle of maximal length.

Proposition 10.1. *Let p and q be coprime positive integers, and let F_1 and F_2 be cyclic permutations of $\mathbb{Z}_p$ and $\mathbb{Z}_q$ respectively. Then $F_1 \times F_2$ is a cyclic permutation of $\mathbb{Z}_p \times \mathbb{Z}_q$ with period pq.*

Proof. The order of (F_1, F_2) in $\mathbb{Z}_p \times \mathbb{Z}_q$ is $\operatorname{lcm}(p, q) = pq$ since $\gcd(p, q) = 1$. Therefore the orbit of any element has size $pq = |\mathbb{Z}_p \times \mathbb{Z}_q|$, so (F_1, F_2) is a single cyclic permutation. □

10.3 Comparison of Periods

Machine	Period	Approximate Magnitude
Enigma (3-rotor)	â‰ˆ 16,900	$\sim 10^{4.2}$
Purple (twenties)	15,625	$\sim 10^{4.2}$
Violet (canonical: r=5, k=6)	$26^5 \cdot 25^6$	$\sim 2.86 \times 10^{15}$

Violet's period exceeds Enigma's and Purple's by more than eleven orders of magnitude.

---

11. Keyspace Analysis and Entropy Calculation

11.1 Key Components

The secret key of the Violet machine comprises several independent components:

1. **Rotor selection and ordering**: Choose r rotors from a set of n available rotors and arrange them in order. This yields $n!/(n-r)!$ possibilities.
2. **Initial rotor positions**: Each of the r rotors can start in any of 26 positions, yielding 26^r possibilities.
3. **Plugboard configuration**: The plugboard implements an involution with m transposition pairs, yielding $I(26, m) = 26!/((26-2m)!\,2^m\,m!)$ possibilities.
4. **Initial switch positions**: Each of the k switches can start in any of 25 positions, yielding 25^k possibilities.

11.2 Total Keyspace

Theorem 11.1 (Total Keyspace). *The total number of distinct key settings for a Violet machine with n available rotors, r installed rotors, m plugboard pairs, and k stepping switches is:*

$$K_{\text{total}} = \frac{n!}{(n-r)!} \cdot 26^r \cdot \frac{26!}{(26-2m)!\,2^m\,m!} \cdot 25^k$$

Proof. Each key component is chosen independently. The rotor selection and ordering gives $n!/(n-r)!$ choices. The initial rotor positions give 26^r choices. The plugboard gives $I(26, m)$ choices. The initial switch positions give 25^k choices. By the multiplication principle, the total is their product. □

11.3 Canonical Machine Instance Computation

We now compute the keyspace for a **canonical Violet machine** with the following parameters:

- $n = 8$ available rotors (rotor inventory)
- $r = 5$ installed rotors
- $k = 6$ stepping switches
- $m = 10$ plugboard pairs

Step 1: Rotor selection and ordering.

$$N_{\text{order}} = \frac{8!}{(8-5)!} = \frac{8!}{3!} = \frac{40320}{6} = 6720$$

$$\log_2(6720) \approx 12.71 \text{ bits}$$

Step 2: Initial rotor positions.

$$N_{\text{pos}} = 26^5 = 11{,}881{,}376$$

$$\log_2(26^5) = 5\log_2(26) \approx 5 \times 4.700 = 23.50 \text{ bits}$$

Step 3: Plugboard configuration.

$$N_{\text{plug}} = \frac{26!}{6!\,2^{10}\,10!} = \frac{26!}{720 \cdot 1024 \cdot 3{,}628{,}800}$$

Computing: $26! = 403{,}291{,}461{,}126{,}605{,}635{,}584{,}000{,}000$

Denominator: $720 \times 1024 \times 3{,}628{,}800 = 2{,}675{,}318{,}169{,}600$

$$N_{\text{plug}} = \frac{403{,}291{,}461{,}126{,}605{,}635{,}584{,}000{,}000}{2{,}675{,}318{,}169{,}600} = 150{,}738{,}274{,}937{,}250$$

$$\log_2(N_{\text{plug}}) \approx 47.10 \text{ bits}$$

Step 4: Initial switch positions.

$$N_{\text{switch}} = 25^6 = 244{,}140{,}625$$

$$\log_2(25^6) = 6\log_2(25) \approx 6 \times 4.644 = 27.86 \text{ bits}$$

Step 5: Total keyspace.

$$K_{\text{total}} = 6720 \times 11{,}881{,}376 \times 150{,}738{,}274{,}937{,}250 \times 244{,}140{,}625$$

To compute the entropy:

$$H(K) = \log_2(K_{\text{total}}) = 12.71 + 23.50 + 47.10 + 27.86 \approx 111.17 \text{ bits}$$

If we also consider that the rotor wirings and switch wiring tables themselves are part of the key (as they would be if the wirings are secret, analogous to the Enigma's rotor wiring being initially unknown to the Allies), the entropy increases dramatically. Each rotor wiring is a permutation in S_{26}, contributing $\log_2(26!) \approx 88.38$ bits per rotor, and each switch position wiring contributes similarly. With $r = 5$ rotors and $k \times 25 = 150$ switch position wirings:

$$H_{\text{wiring}} = 5 \times 88.38 + 150 \times 88.38 \approx 441.9 + 13{,}257 = 13{,}698.9 \text{ bits}$$

However, in practice, the wiring tables would be fixed for a given machine type (as with the Enigma), so the **operational keyspace** - the entropy associated with settings that change daily or per-message - is approximately **111 bits**. If we consider the wiring as secret (as in an initial cryptanalytic attack where the machine design is unknown), the total entropy exceeds **13,800 bits**.

11.4 Comparison of Keyspace Sizes

Machine	Operational Keyspace	Entropy (bits)
Enigma (3-rotor, 10-pair plugboard)	$\approx 1.59 \times 10^{23}$	≈ 77
Enigma (4-rotor, M4 variant)	$\approx 10^{26}$	≈ 86
Purple	$\approx 10^{18}$	≈ 60
Violet (canonical, operational)	$\approx 2^{111}$	≈ 111
Violet (including secret wirings)	$\approx 2^{13800}$	$\approx 13{,}800$

The canonical Violet configuration provides approximately 111 bits of operational key entropy, exceeding the three-rotor Enigma by approximately 34 bits (a factor of $\sim 10^{10}$) and Purple by approximately 51 bits (a factor of $\sim 10^{15}$).

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12. Statistical Properties and Mixing Behaviour

12.1 Mixing on Symmetric Groups

The statistical behaviour of the Violet cipher is best understood through the theory of mixing on finite groups. Consider the sequence of permutations $\{E_t\}_{t=0}^{N-1}$ produced by one full period of the machine. We ask: how closely does the empirical distribution of this sequence approximate the uniform distribution on S_{26}?

For a single rotor (without plugboard), the 26 permutations $\{R(0), R(1), \ldots, R(25)\}$ form a coset of the cyclic group $\langle \tau_1 \rangle$ in S_{26}, which is a very thin slice of S_{26}. However, when multiple rotors are composed, the product $R_r(t) \circ \cdots \circ R_1(t)$ ranges over a much larger subset of S_{26}, and the further composition with the stepping-switch permutation σ_t provides additional mixing.

12.2 The Diaconis-Shahshahani Framework

Diaconis and Shahshahani (1981) showed that for random transpositions on S_n, the mixing time is $\Theta(n \log n)$. More generally, the product of $\Omega(n \log n)$ independently and uniformly chosen permutations from a generating set converges rapidly to the uniform distribution on S_n in total variation distance.

While the Violet cipher does not produce independent random permutations (the sequence is deterministic once the key is set), the large number of independently wired components (r rotors Ã— 26 positions + k switches Ã— 25 positions = $r \cdot 26 + k \cdot 25$ distinct wiring tables) ensures that the achievable permutations span a large subset of S_{26}, and the time-varying composition produces effective mixing.

12.3 Deviation from Uniformity

Define the **total variation distance** between the empirical distribution of $\{E_t\}$ and the uniform distribution on S_{26}:

$$d_{TV} = \frac{1}{2} \sum_{\pi \in S_{26}} \left| \frac{|\{t : E_t = \pi\}|}{N} - \frac{1}{26!} \right|.$$

For a well-designed Violet machine (with independently and quasi-randomly chosen wiring tables), we expect d_{TV} to be small, though it cannot be exactly zero since $N = 26^r \cdot 25^k$ is finite and much smaller than $26!$ for practical parameter choices. The key observation is that even though the machine cannot produce all $26!$ permutations (since $N \ll 26!$ for practical parameters), the permutations it does produce are well-distributed in the sense that for any fixed input x, the outputs $\{E_t(x)\}_{t=0}^{N-1}$ are approximately equidistributed over Σ.

12.4 Flat Output Distribution

Proposition 12.1. *For the rotor stage alone, with r rotors and generic wiring choices, the distribution of $\rho_t(x)$ over one full rotor period $N_\rho = 26^r$ is exactly uniform over Σ for each fixed x.*

Proof. For each fixed x and each target $y \in \Sigma$, write $\rho_t(x) = y$ as a constraint on the rotor state vector. As the fastest rotor (rotor 1) cycles through all 26 positions with all other rotors fixed, the output $\rho_t(x)$ takes each value in Σ exactly once (since $R_1(t)$ is a permutation for each t). Summing over all configurations of the remaining rotors preserves this uniformity. Therefore, out of 26^r total states, exactly 26^{r-1} map x to any particular y, giving $\Pr(\rho_t(x) = y) = 26^{r-1}/26^r = 1/26$. $\square$

This is a strong flatness property that holds regardless of the specific wiring choices. The composition with σ_t preserves this property (since σ_t is a bijection at each t), so the full cipher also exhibits flat output distribution over a full period.

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13. Cryptanalytic Considerations

This section provides a detailed assessment of the resistance of the Violet cipher to the principal classes of attacks that have been applied to historical electromechanical cipher machines. We consider attacks in increasing order of sophistication.

13.1 Ciphertext-Only Attacks

In a ciphertext-only attack, the adversary possesses only the ciphertext $z_0, z_1, \ldots, z_{L-1}$ and seeks to recover the plaintext or the key. For a time-varying polyalphabetic cipher, the primary ciphertext-only technique is **frequency analysis** applied to individual cipher alphabets.

If the attacker knows (or can estimate) the machine period N, characters at positions $t, t+N, t+2N, \ldots$ are all encrypted under the same permutation E_t and can be subjected to monalphabetic frequency analysis. However, for the canonical Violet instance with $N \approx 2.86 \times 10^{15}$, accumulating even two characters encrypted under the same permutation requires a ciphertext of length exceeding 10^{15} characters - far beyond any practical message volume. Therefore, **period-based frequency analysis is infeasible against Violet**.

Without period exploitation, the attacker must contend with a polyalphabetic cipher of effective period N, which is equivalent to a stream of N distinct monalphabetic substitutions. Standard statistical attacks (such as the Kasiski test or index of coincidence) provide no useful information when the period exceeds the message length, which will invariably be the case in practice.

13.2 Known-Plaintext Attacks

In a known-plaintext attack (KPA), the adversary possesses a set of plaintext-ciphertext pairs (x_t, z_t) for $t \in T$, where T is a known set of time indices. From each pair, the attacker learns the value $E_t(x_t) = z_t$, which provides one constraint on the permutation E_t. Since E_t has 26 entries, a complete determination of E_t requires 26 known-plaintext pairs at the same time step - which is impossible in a stream cipher (only one character is encrypted per time step). Therefore, each known-plaintext pair provides only **one bit of information** about the corresponding permutation E_t.

The attacker's task is to find a key K (comprising rotor selection, rotor positions, plugboard, and switch positions) such that the resulting sequence of permutations $\{E_t^{(K)}\}$ is consistent with all observed pairs. For the canonical instance with $\sim 2^{111}$ possible keys, and each known-plaintext pair eliminating approximately a $1 - 1/26$ fraction of keys, the attacker needs approximately $111 / \log_2(26) \approx 24$ known-plaintext pairs to reduce the expected number of surviving keys to 1. However, **finding** the correct key among 2^{111} candidates using these constraints requires an efficient search algorithm, which we now discuss.

13.3 Bombe-Style Crib Attacks

The most effective attack against the wartime Enigma was the Bombe, developed by Rejewski, Turing, and Welchman. The Bombe exploited the structure of the Enigma to test candidate key settings against a known or guessed plaintext fragment (a "crib"). Its effectiveness relied on several structural features of the Enigma:

1. **Reciprocity**: $E_t = E_t^{-1}$, allowing the construction of closed loops in the crib-ciphertext pairing graph.

2. **Self-stenciling**: $E_t(x) \neq x$, used to eliminate candidate crib positions.
3. **Plugboard structure**: The plugboard permutation P commutes with the rotor stack in a specific way (acting at both ends of the signal path), enabling the separation of plugboard and rotor hypotheses.
4. **Small rotor state space**: Only $26^3 \approx 17{,}576$ rotor states need be tested.

Violet negates **all four** of these structural features:

1. **No reciprocity**: $E_t \neq E_t^{-1}$ in general, so closed loops in the crib graph cannot be formed.
2. **No self-stenciling guarantee**: $E_t(x) = x$ is possible, so plaintext-ciphertext identity cannot be used to eliminate crib positions.
3. **No plugboard separation**: The plugboard feeds only into the rotor stage, and its effect is entangled with the rotor and switch permutations in a non-separable way. The cascade with the switch stage destroys the bilateral symmetry that enabled Welchman's diagonal board.
4. **Large state space**: With $26^5 \cdot 25^6 \approx 2.86 \times 10^{15}$ states, exhaustive testing is infeasible.

Conclusion: Bombe-style crib attacks, as developed against the Enigma, do not directly generalize to the Violet cipher.

13.4 Depth Attacks

A **depth** occurs when two or more messages are encrypted with the same key setting (i.e., the same initial state). In this case, the same sequence of permutations $\{E_t\}$ is applied to different plaintexts, and the differences between the ciphertexts reflect differences between the plaintexts:

$$z_t - z'_t = E_t(x_t) - E_t(x'_t)$$

For a general permutation E_t, this difference is not simply $x_t - x'_t$ (as it would be for an additive cipher), but it nonetheless provides structural information about E_t that can be exploited, particularly when combined with language statistics.

Depth attacks against Violet would proceed similarly to depth attacks against the Enigma, exploiting the redundancy in natural language plaintext. However, the absence of reciprocity and self-stenciling removes two powerful tools, making depth attacks against Violet less efficient than against the Enigma. The primary defense against depth attacks is operational: enforce unique message keys, as was standard practice with the Enigma's message key system (albeit poorly implemented in practice).

13.5 Cycle-Structure Attacks

Permutation cycle-structure attacks exploit the fact that the cycle decomposition of a permutation is a conjugacy-class invariant: if $E_t = \alpha \circ \beta$, then the cycle structure of E_t constrains the possible factorizations. For the Violet cipher, observing the cycle structure of E_t (which requires complete knowledge of all 26 input-output pairs at time t, i.e., 26 known-plaintext characters at the same position) could in principle constrain the joint cycle structures of ρ_t and σ_t.

However, as noted above, obtaining 26 simultaneous known-plaintext pairs is impossible in a stream cipher context. Moreover, even with knowledge of E_t's cycle structure, the number of factorizations $E_t = \sigma \circ \rho$ is astronomically large (of order $(26!)^2/|S_{26}| = 26!$ by elementary counting), and without additional structural constraints, this information is insufficient to recover the individual stages.

13.6 Meet-in-the-Middle Attacks

The cascade structure $E_t = \sigma_t \circ \rho_t$ invites meet-in-the-middle (MITM) attacks. In a standard MITM attack on a two-stage cascade, the attacker:

1. Enumerates all possible first-stage outputs $y = \rho_t(x)$ for each candidate rotor key.

2. Enumerates all possible second-stage inputs $y = \sigma_t^{-1}(z)$ for each candidate switch key.
3. Searches for matches $\rho_t(x) = \sigma_t^{-1}(z)$.

Theorem 13.1 (MITM Complexity). *A meet-in-the-middle attack on the Violet cipher requires:*

- *Time complexity: $O(|\mathcal{K}_\rho| + |\mathcal{K}_\sigma|)$ where $|\mathcal{K}_\rho|$ and $|\mathcal{K}_\sigma|$ are the rotor and switch keyspace sizes respectively.*
- *Space complexity: $O(\min(|\mathcal{K}_\rho|, |\mathcal{K}_\sigma|))$ for storing the intermediate values.*

For the canonical instance:

$$|\mathcal{K}_\rho| = N_{\text{order}} \cdot N_{\text{pos}} \cdot N_{\text{plug}} = 6720 \times 26^5 \times 150{,}738{,}274{,}937{,}250 \approx 2^{83.3}$$

$$|\mathcal{K}_\sigma| = N_{\text{switch}} = 25^6 \approx 2^{27.9}$$

The MITM attack requires time $O(2^{83.3})$ and space $O(2^{27.9})$.

Proof. The MITM attack proceeds by choosing the intermediate point $y = \rho_t(x_t)$ as the meeting value. For each candidate rotor key, compute $y = \rho_t(x_t)$ and store it in a lookup table indexed by y. For each candidate switch key, compute $y' = \sigma_t^{-1}(z_t)$ and look up y' in the table. A match indicates a candidate key pair. The time is dominated by the larger enumeration, and the space by the smaller table. □

Remark. While $2^{83.3}$ is a significant reduction from the brute-force 2^{111}, it remains astronomically beyond the reach of 1940s technology and indeed of any currently foreseeable technology. For comparison, the entire Bitcoin network performs approximately 2^{67} hash operations per year.

However, from a theoretical perspective, the MITM attack demonstrates that the cascade does not provide security equivalent to the product of the individual keyspaces, but rather the sum of their logarithms. This is a well-known principle in cascade cipher design and motivates making both stages independently strong.

13.7 Why Cascading Increases Attack Complexity

The fundamental reason that the cascade resists cryptanalysis can be summarized as follows:

1. **No structural shortcuts**: The rotor and switch stages use fundamentally different mechanisms to generate permutations (conjugation versus table lookup), and there is no known algebraic relationship between the two families of permutations that would enable joint analysis.
2. **Independent evolution**: The rotor and switch states evolve independently, so knowledge of one stage's state at time t provides no information about the other stage's state.
3. **Non-commutative composition**: Since S_{26} is non-abelian, knowledge of the composite $E_t = \sigma_t \circ \rho_t$ does not uniquely determine ρ_t and σ_t, even if E_t is completely known.
4. **Coprime periods**: The coprimality of N_ρ and N_σ ensures that the relative alignment of the rotor and switch states shifts continuously and does not repeat until the full period $N = N_\rho \cdot N_\sigma$ has elapsed, maximizing the number of distinct composite permutations.

14. Comparison With Historical Machines

14.1 Comparative Summary Table

Property	Enigma (3-rotor)	Purple	Violet (canonical)
Architecture	Rotor + reflector	Stepping switches	Rotor + stepping switch cascade
Alphabet treatment	Full 26 letters	Split: 6 + 20	Full 26 letters
Reciprocal?	Yes ($E_t = E_t^{-1}$)	No	No
Self-stenciling?	Yes ($E_t(x) \neq x$)	Possible	Possible
Reflector?	Yes	No	No
Number of rotors	3 (from 5)	0	5 (from 8)
Number of switches	0	4 (one 6-level, three 20-level)	6
Plugboard	10 pairs	None	10 pairs
Operational keyspace	$\approx 2^{77}$	$\approx 2^{60}$	$\approx 2^{111}$
Period	$\approx$ 16,900	$15{,}625$ (twenties)	$\approx 2.86 \times 10^{15}$
Bombe-vulnerable?	Yes	N/A	No

Property	Enigma (3-rotor)	Purple	Violet (canonical)
Alphabet partition?	No	Yes (critical weakness)	No
MITM complexity	N/A (single stage)	N/A	$O(2^{83})$

14.2 Why Violet is Stronger than Enigma

The Violet design addresses each of Enigma's identified weaknesses:

1. **Reflector elimination**: By removing the reflector, Violet eliminates the involutory constraint and the self-stenciling property, which were the two most important entry points for Bletchley Park's attacks.
2. **Cascade architecture**: The addition of the stepping-switch stage acts as a cryptographic multiplier, composing the rotor permutation with an independently evolving second permutation.
3. **Larger state space**: With five rotors and six switches, Violet has $26^5 \cdot 25^6 \approx 2.86 \times 10^{15}$ internal states, compared to Enigma's $\sim 17{,}000$.
4. **Longer period**: The coprimality of the rotor and switch periods ensures a period that is the product (not just the LCM) of the individual periods.
5. **No structural shortcuts**: The two-stage cascade admits no equivalent of the Bombe, which relied critically on the Enigma's algebraic structure.

14.3 Why Violet is Stronger than Purple

1. **No alphabet partition**: Purple's fatal weakness - the fixed partition of the alphabet into sixes and twenties - is absent from Violet, which processes all 26 letters through both stages.
2. **Additional rotor stage**: Purple used only stepping switches; Violet adds a rotor stage with its own keyspace contribution.

3. **Larger keyspace**: Violet's $\sim 2^{111}$ operational keyspace exceeds Purple's $\sim 2^{60}$ by a factor of $\sim 2^{51}$.

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15. Mechanical Feasibility and Engineering Considerations

15.1 Historical Technology Base

The Violet cipher, while hypothetical, is designed to be fully realizable using electromechanical technologies available in the late 1930s and 1940s. We describe how each component maps to existing hardware.

15.2 Rotor Assembly

The rotor stage is directly derived from established rotor machine technology. Each rotor is a disk of insulating material (Bakelite or hard rubber) with 26 electrical contacts on each face, connected by internal wiring. The rotor assembly would be physically similar to the Enigma's, albeit without the reflector and with five rotors instead of three. The odometer stepping mechanism - pawl-and-ratchet or Geneva drive - was well understood and widely implemented.

The omission of the reflector actually simplifies the mechanical design: the signal path is unidirectional through the rotor stack, eliminating the need for the return path and the reflector drum. The machine would require separate encryption and decryption modes (achieved by reversing the rotor stack electrically or by using precomputed inverse wiring tables), introducing slight operational complexity but no mechanical difficulty.

15.3 Stepping-Switch Assembly

The stepping-switch stage employs standard **Strowger stepping switches** (also called uniselector switches), which were the backbone of automatic telephone exchanges from the 1890s through the mid-twentieth century. A 25-point uniselector has a rotating wiper arm that makes contact with 25 fixed terminals in sequence. By wiring 26 input lines to the wiper contacts through 26 cross-connections at each of the 25 positions, a single switch implements a sequence of 25 permutations of the 26-letter alphabet as it cycles through its positions.

The cascaded stepping mechanism (switch 1 drives the advancement of switch 2 upon completing a full cycle) is identical in principle to the cascaded decade switches used in automatic telephone exchanges and was a standard design pattern. The six-switch cascade required for the canonical Violet instance would occupy approximately 30 cm Ã— 30 cm Ã— 40 cm and weigh roughly 5-10 kg, well within the size and weight budgets of military cipher equipment.

15.4 Relay Logic

The interconnection between the rotor stage and the stepping-switch stage requires routing 26 electrical lines from the rotor output to the switch input. This is accomplished by simple hard-wiring; no active components are needed. However, signal regeneration may be necessary if the total resistance of the rotor-switch cascade causes excessive voltage drop. Standard telephone relays (operating on 24-50 V DC, with switching times of 5-20 ms) would provide adequate signal amplification and isolation.

A complete Violet machine would require approximately 52 relays for signal routing (26 for rotor-to-switch interconnection, 26 for switch-to-output), plus additional relays for stepping logic (approximately 20-30 relays for the odometer carry chain and switch cascade logic). The total relay count of approximately 80-100 is comparable to the relay count in the Purple machine (which used approximately 100 relays) and well within the manufacturing capabilities of the era.

15.5 Power and Timing

Each encryption cycle involves: (1) advancing the rotor and switch states, (2) closing the keyboard contact, (3) propagating the electrical signal through the plugboard, rotor stack, and switch cascade, and (4) illuminating the output lamp. With relay switching times of 10-20 ms and signal propagation through passive wiring essentially instantaneous, a complete encryption cycle could be completed in approximately 50-100 ms, allowing an operator typing speed of 10-20 characters per second. This is comparable to the Enigma's operational speed.

Power requirements would be a 24 V or 48 V DC supply (standard for telephone equipment), with total power consumption of approximately 20-50 W - easily supplied by batteries for field use or by mains power for fixed installations.

15.6 Size and Weight Estimate

Component	Approximate Size	Approximate Weight
Rotor assembly (5 rotors)	15 Ã— 15 Ã— 20 cm	3 kg
Plugboard	20 Ã— 15 Ã— 5 cm	1 kg
Stepping switches (6 units)	30 Ã— 30 Ã— 40 cm	8 kg
Relay chassis	20 Ã— 15 Ã— 15 cm	3 kg
Keyboard and lampboard	30 Ã— 25 Ã— 5 cm	2 kg
Power supply	15 Ã— 10 Ã— 10 cm	3 kg
Housing and chassis	-	5 kg
Total	**~50 Ã— 40 Ã— 45 cm**	**~25 kg**

This places the Violet machine in the size class of a large portable field cipher machine - larger than the Enigma (approximately 12 kg) but comparable to the SIGABA (approximately 45 kg) and smaller than the Lorenz SZ40/42 (approximately 55 kg). The machine would be suitable for division-level or higher military communications, fixed diplomatic installations, and naval shore stations.

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16. Security Limitations from a Modern Perspective

16.1 Deterministic Keystream

The Violet cipher, like all classical electromechanical cipher machines, generates a deterministic keystream: once the initial state (key) is set, the sequence of encryption permutations $\{E_t\}$ is entirely determined. There is no source of entropy during encryption. This means that:

1. **Keystream reuse**: If two messages are encrypted with the same key, the attacker obtains a depth, which can be exploited through language statistics.
2. **Chosen-plaintext attacks**: An attacker who can cause the encryption of known plaintext can directly observe portions of the keystream.
3. **No forward secrecy**: Compromise of the key reveals all past and future messages encrypted under that key.

16.2 No Information-Theoretic Security

By Shannon's theory, perfect secrecy requires that the key space be at least as large as the message space. Since practical messages can be arbitrarily long and the Violet key is finite, the system cannot achieve information-theoretic security. This is a limitation shared by all practical cipher systems, classical and modern, except one-time pad systems.

16.3 Vulnerability to Modern Computing

From a modern computational perspective, the operational keyspace of $\sim 2^{111}$ bits, while enormous by 1940s standards, is within the range of contemporary cryptanalytic capabilities for well-funded adversaries, though it remains beyond casual attack. Modern symmetric ciphers (e.g., AES-256 with 256-bit keys) provide substantially larger security margins. The meet-in-the-middle complexity of $\sim 2^{83}$ is a more relevant security measure and would be considered insufficient by modern standards (where 128-bit security is the recommended minimum).

16.4 Side-Channel Vulnerabilities

Electromechanical devices are inherently vulnerable to side-channel attacks:

- **Acoustic emanations**: The clicking of relays and stepping switches could reveal information about the machine's internal state.
- **Electromagnetic emissions**: Current flow through the rotor and switch wiring produces detectable electromagnetic radiation (an early form of what would later be called TEMPEST).
- **Timing variations**: Differences in relay switching times could leak state information.

These side channels, while theoretical in the 1940s context, represent genuine vulnerabilities that would be relevant against a modern adversary.

16.5 Algebraic Attacks

Modern algebraic cryptanalysis techniques - including GrÃ¶bner basis methods, SAT-solver attacks, and constraint satisfaction approaches - could potentially be applied to the Violet cipher's algebraic structure. Given enough known plaintext-ciphertext pairs, the problem of recovering the key can be formulated as a system of polynomial equations over $\mathbb{Z}_{26}$ (or, more precisely, as a constraint satisfaction problem over finite domains), and modern solvers might find solutions more efficiently than brute-force search suggests. A detailed analysis of algebraic attacks is beyond the scope of this paper but represents an important direction for future work.

17. Conclusion

This paper has presented a rigorous mathematical and cryptographic analysis of the Violet cipher, a cascade electromechanical cipher machine that combines a reflectorless rotor permutation stage with an independent stepping-switch permutation stage. The key findings are:

1. **Closure and well-definedness**: The Violet encryption function is a well-defined permutation of the 26-letter alphabet at every time step (Theorem 9.1).
2. **Non-reciprocity**: Unlike the Enigma, the Violet cipher is not its own inverse, eliminating the self-stenciling property that was critical to the Enigma's defeat (Theorem 9.2).
3. **Maximal period**: The coprimality of the rotor and switch cycle lengths ensures that the machine period equals the product of the individual periods, yielding a period of $\sim 2.86 \times 10^{15}$ for the canonical instance - over eleven orders of magnitude longer than Enigma or Purple (Theorem 9.5).
4. **Large keyspace**: The canonical instance provides approximately 111 bits of operational key entropy, exceeding the Enigma by 34 bits and Purple by 51 bits (Theorem 11.1).
5. **Resistance to classical attacks**: The cascade architecture resists Bombe-style crib attacks, cycle-structure attacks, and depth attacks, while meet-in-the-middle attacks require $O(2^{83})$ operations (Section 13).
6. **Statistical mixing**: The two-stage composition promotes rapid approach to uniformity in the distribution of output permutations, with the deviation from uniformity decreasing as the number of independent components increases (Theorem 9.7).
7. **Mechanical feasibility**: The design is fully realizable using mid-twentieth-century electromechanical technology, with an estimated machine weight of approximately 25 kg and encryption speed of 10-20 characters per second (Section 15).

The Violet cipher represents a historically plausible "what-if" design that, had it been conceived and implemented in the 1940s, would have posed substantially greater challenges to cryptanalysts than the machines actually deployed during the war. Its study illuminates the cryptographic consequences of cascade composition and provides a concrete example of how the combination of independent permutation-generating mechanisms can yield a system whose security exceeds that of either mechanism alone.

Future work could include: detailed algebraic cryptanalysis using modern computational tools; construction of a physical or simulated prototype; analysis of irregular stepping mechanisms (analogous to SIGABA); and extension of the framework to larger alphabets or non-cyclic stepping patterns.

Appendix A - Fixed Point Analysis

A.1 Expected Number of Fixed Points

For a permutation π chosen uniformly at random from S_n, the expected number of fixed points is:

$$\mathbb{E}[X] = \sum_{i=0}^{n-1} \Pr(\pi(i) = i) = \sum_{i=0}^{n-1} \frac{1}{n} = 1.$$

This result holds for all $n \geq 1$ and is independent of n.

A.2 Distribution of Fixed Points

The number of permutations in S_n with exactly j fixed points is:

$$f(n, j) = \binom{n}{j} D_{n-j}$$

where D_m is the number of derangements (fixed-point-free permutations) of m elements:

$$D_m = m! \sum_{i=0}^{m} \frac{(-1)^i}{i!}.$$

The probability of exactly j fixed points is:

$$\Pr(X = j) = \frac{f(n, j)}{n!} = \frac{1}{j!} \sum_{i=0}^{n-j} \frac{(-1)^i}{i!} \xrightarrow{n \to \infty} \frac{e^{-1}}{j!}.$$

For $n = 26$:

j	$\Pr(X = j)$
0	0.3679
1	0.3679
2	0.1839
3	0.0613
4	0.0153
5	0.0031
≥ 6	0.0006

The probability that the Violet cipher encrypts **at least one** letter to itself in a given alphabet is $1 - \Pr(X = 0) \approx 0.6321 = 63.21\%$. In contrast, this probability is exactly 0% for the Enigma due to its reflector.

A.3 Cryptographic Implications

The existence of fixed points in Violet means that the cipher does not possess the Enigma's "self-stenciling" property ($E_t(x) \neq x$). While self-stenciling was operationally useful for the Enigma (it allowed checking whether a decryption attempt was incorrect by the presence of identity mappings), it was also a devastating weakness, as it allowed Bletchley Park to eliminate incorrect crib positions. The absence of self-stenciling in Violet removes this avenue of attack.

Appendix B - Group Theory of the Symmetric Group

B.1 Basic Properties

The symmetric group S_n satisfies:

- **Order**: $|S_n| = n!$
- **Non-abelian** for $n \geq 3$: $\exists\, \alpha, \beta \in S_n$ such that $\alpha\beta \neq \beta\alpha$
- **Generated by transpositions**: Every permutation is a product of transpositions
- **Generated by adjacent transpositions**: $S_n = \langle (1\,2), (2\,3), \ldots, (n{-}1\;n) \rangle$
- **Generated by a transposition and an n-cycle**: $S_n = \langle (1\,2), (1\,2\,\cdots\,n) \rangle$

B.2 Conjugacy Classes

Two permutations $\alpha, \beta \in S_n$ are conjugate ($\alpha \sim \beta$) if and only if they have the same cycle type. The number of conjugacy classes of S_n equals the number of partitions of n, denoted $p(n)$. For $n = 26$, $p(26) = 2{,}436$.

B.3 Relevance to Cipher Design

The non-commutativity of S_{26} is fundamental to cascade cipher security: since $\sigma \circ \rho \neq \rho \circ \sigma$ in general, knowledge of $E = \sigma \circ \rho$ does not allow the attacker to separate the contributions of σ and ρ without additional information. The number of factorizations of a given $E \in S_{26}$ as $E = \sigma \circ \rho$ with $\sigma, \rho \in S_{26}$ is exactly $|S_{26}| = 26!$, since for each choice of ρ, the value $\sigma = E \circ \rho^{-1}$ is uniquely determined.

Appendix C - Detailed Keyspace Calculations

C.1 Plugboard Combinations

For $m = 10$ pairs from 26 letters:

$$I(26, 10) = \frac{26!}{6! \cdot 2^{10} \cdot 10!}$$

Step-by-step:

- $26! = 403{,}291{,}461{,}126{,}605{,}635{,}584{,}000{,}000$
- $6! = 720$
- $2^{10} = 1{,}024$
- $10! = 3{,}628{,}800$
- Denominator: $720 \times 1{,}024 \times 3{,}628{,}800 = 2{,}675{,}318{,}169{,}600$
- $I(26, 10) = 150{,}738{,}274{,}937{,}250$

C.2 Complete Keyspace Arithmetic

$$K = 6{,}720 \times 11{,}881{,}376 \times 150{,}738{,}274{,}937{,}250 \times 244{,}140{,}625$$

Computing step by step:

- $A = 6{,}720 \times 11{,}881{,}376 = 79{,}842{,}846{,}720$
- $B = A \times 150{,}738{,}274{,}937{,}250 \approx 1.203 \times 10^{25}$
- $K = B \times 244{,}140{,}625 \approx 2.938 \times 10^{33}$

$$H(K) = \log_2(K) \approx \log_2(2.938 \times 10^{33}) = 33 \times \log_2(10) + \log_2(2.938)$$

$$\approx 33 \times 3.3219 + 1.555 \approx 109.6 + 1.6 \approx 111.2 \text{ bits}$$

C.3 Machine Period Calculation

$$N = 26^5 \times 25^6 = 11{,}881{,}376 \times 244{,}140{,}625 = 2{,}900{,}429{,}687{,}500{,}000{,}000{,}000$$

$$\approx 2.9 \times 10^{15}$$

$$\log_2(N) \approx 51.4 \text{ bits}$$

At a sustained encryption rate of 100 characters per minute (typical for a skilled operator), the machine would require approximately 2.9×10^{13} minutes $\approx$ 55 million years to complete one full period - confirming that period-based attacks are entirely impractical.

Appendix D - Notation Summary

Symbol	Meaning
Σ	Alphabet $\{0, 1, \ldots, 25\}$
S_{26}	Symmetric group on Σ
τ_k	Cyclic shift: $x \mapsto x + k \pmod{26}$
P	Plugboard permutation (involution)
R_i	Base wiring of rotor i
$R_i(t)$	Effective permutation of rotor i at time t
ρ_t	Composite rotor-stage permutation at time t
$W_j^{(p)}$	Permutation of switch j at position p
σ_t	Composite switch-stage permutation at time t
E_t	Full encryption permutation: $E_t = \sigma_t \circ \rho_t$
$\mathbf{s}^{(\rho)}(t)$	Rotor state vector $\in \mathbb{Z}_{26}^r$
$\mathbf{s}^{(\sigma)}(t)$	Switch state vector $\in \mathbb{Z}_{25}^k$
$\mathbf{s}(t)$	Combined state vector
$\mathcal{S}$	State space $\mathbb{Z}_{26}^r \times \mathbb{Z}_{25}^k$
F	State transition function
G	Output function: state $\to$ permutation
N_ρ	Rotor period: 26^r
N_σ	Switch period: 25^k

Symbol	Meaning
N	Machine period: $\mathrm{lcm}(N_\rho, N_\sigma)$
K_{total}	Total keyspace size
$H(K)$	Key entropy: $\log_2(K_{\text{total}})$
r	Number of installed rotors
k	Number of stepping switches
m	Number of plugboard pairs

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